P2 — The PDF files of the project folder: which were used, and how

Provenance required by the task specification (PROJECT spacetimes)

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1. What this document is for

The task specification asks for a provenance that lists every PDF file that was used and says how it was used. The project folder contains nine PDF files, all of them published papers. This document lists them, says how each was read, what the project took from it, where the project cites it, and what it did *not* take. The symbols of these papers, and the places where they collide with the symbols of the notebooks, are the subject of the MASTER provenance (`provenance/P1_master_symbols`, Sec. 5.2).

The result in one sentence: **no formula, no matrix and no number of the project is copied from any of the nine papers**; the papers are used for the origin and the conventions of the author's matrices, for the mathematical background of the two spacetimes, and as references. Everything that the project asserts about its matrices is proved by its own programs (provenance "dirac matrices").

The nine files are not part of the repository, because they are copyrighted publications. They are identified by their SHA-256 checksums (Sec. 5), so that a reader who has the files can verify that he has the same copies.

2. The nine files

"Pages" is the number of pages of the file. "Text" says how the text was obtained: *digital* means that the file is the publisher's digital original; *layer* that it is a scan with a text layer made by character recognition before the project (of mixed quality: words are reliable, Greek letters and formulas often are not); *OCR* that it is a scan without text, which was read by optical character recognition during this project (program tesseract 5.3.4) with the same limits.

Id	File	Paper	Pages	Text
D01	`6dim dirac.pdf`	E. A. Lord, "The Dirac spinor in six dimensions", Proc. Camb. Phil. Soc. 64, 765 (1968)	14	OCR
D02	`A Remarkable Representation.pdf`	P. A. M. Dirac, "A remarkable representation of the 3 + 2 de Sitter group", J. Math. Phys. 4, 901 (1963)	10	OCR
D03	`A new spin-half.pdf`	P. L. Nash, "A new spin-1/2 wave equation", J. Math. Phys. 26, 1363 (1985)	5	OCR
D04	`Bryant-octon-17-2-185-txt.pdf`	R. L. Bryant, "Submanifolds and special structures on the octonians", J. Differential Geom. 17, 185 (1982)	48	layer
D05	`Calibrated geometries.pdf`	R. Harvey and H. B. Lawson, Jr., "Calibrated geometries", Acta Math. 148, 47 (1982)	111	layer
D06	`Identities satisfied.pdf`	P. L. Nash, "Identities satisfied by the generators of the Dirac algebra", J. Math. Phys. 25, 204 (1984)	6	OCR

Id	File	Paper	Pages	Text
D07	`On the structure of the split octonion algebra.pdf`	P. L. Nash, “On the structure of the split octonion algebra”, Nuovo Cimento B 105, 31 (1990)	11	layer
D08	`Second gravity.pdf`	P. L. Nash, “Second gravity”, J. Math. Phys. 51, 042501 (2010)	27	digital
D09	`Vector cross products.pdf`	R. B. Brown and A. Gray, “Vector cross products”, Comment. Math. Helv. 42, 222 (1967)	15	layer

Because of the limits of the recognized text, the project relies only on statements of these papers that are legible *in words* (for example “the tilde denotes transpose”, or the ranges of the indices). Where a formula could not be read with certainty, this is said (Sec. 3, D07).

3. How each paper was used

D08, “Second gravity” (Nash 2010). *Read:* the passages that define the space $R^{4,4}$ and the matrices (its Sec. II), and the list of references. *Used for:* (i) the origin of the matrices of `4+4.nb`: the paper defines real 8×8 matrices τ^A, $\bar{\tau}^A$ with $\tau^A\bar{\tau}^B + \tau^B\bar{\tau}^A = 2G_{AB}$ and the 16×16 matrices t^A with the blocks $(0, \bar{\tau}^A; \tau^A, 0)$, which is the construction of the notebook; (ii) the conventions that differ from the notebook: the index runs from 1 to 8 with $\tau^8 = \bar{\tau}^8 = 1$, the metric has the diagonal $(1,1,1,-1,-1,-1,-1,1)$, a tilde denotes the transpose; (iii) the statement that the construction follows a procedure of Lord (D01). *Not used:* its matrices themselves (the project takes the matrices from the notebook and did not compare them entry by entry with the paper), and its physics (a theory of gravity). *Cited:* both manuscripts, Secs. I and II; P1, Secs. 5.2 and 6.

D07, “On the structure of the split octonion algebra” (Nash 1990). *Read:* the passages that define the matrices (its Sec. 2). *Used for:* the same definitions as D08, in the same conventions; its remark that, relative to an earlier paper of the author, “tau bar” and “tau” are exchanged and the sign of τ^7 is changed, which is a warning for anyone who compares the papers with one another; its condition that a product of the seven matrices with $A = 1, \ldots, 7$ be $+1$. In the scanned copy the formula of that product is not legible enough to say which factors carry a bar; the project therefore states its own result, $\tau^1_{1644} \cdots \tau^7_{1644} = 1$ (check `44-B1`), and does not claim that this is the convention of the paper. *Not used:* the relation to the split octonion algebra and to triality; the manuscripts refer to the paper for it and say so (“not established here”). *Cited:* both manuscripts, Secs. I and II; P1.

D06, “Identities satisfied by the generators of the Dirac algebra” (Nash 1984). *Read:* the passages that define its matrices (its Sec. 2). *Used for:* the origin of the six antisymmetric 4×4 matrices s_h, t_h from which the construction starts: the paper calls the six antisymmetric ones among its real 4×4 matrices s^h (self-dual) and t^h (anti-self-dual), and the notebooks keep these letters (`s4by4[h]`, `t4by4[h]`). *Not used:* its identities, which concern four-dimensional matrices. *Cited:* manuscripts, Sec. I and (4+4) Sec. II A; P1.

D01, “The Dirac spinor in six dimensions” (Lord 1968). *Read:* its introduction and the passages on the generators. *Used for:* one statement only: D08 says that its construction follows the procedure of this paper, which builds spinor representations of orthogonal groups from the generators of a Dirac algebra of lower dimension. *Cited:* manuscripts, Sec. I.

D02, “A remarkable representation of the 3 + 2 de Sitter group” (Dirac 1963). *Read:* introduction. *Used for:* the lineage of the real representations. The notebook `Dirac_Einstein-Lovelock_4+4_Nash.nb` says that the calculation “has its roots” in this paper, in which Dirac defines a real representation of the covering group of $SO(3,2)$, and that the author extended it to $SO(3,3)$, to $SO(4,4)$ and to real 16×16 matrices. The Introduction of the manuscripts repeats this lineage in one sentence. *Cited:* manuscripts, Sec. I.

D03, “A new spin-1/2 wave equation” (Nash 1985). *Read:* abstract and introduction. *Used for:*

the same lineage (the stage of $SO(3,3)$), together with D06. *Not used:* its wave equation. *Cited:* manuscripts, Sec. I.

D04 (Bryant 1982), D05 (Harvey and Lawson 1982), D09 (Brown and Gray 1967). *Read:* their introductions. *Used for:* the statement that the signature 7+1 is the case of the octonions, and as references for the algebra of the octonions, the groups G_2 and $Spin(7)$ and the vector cross products of seven and eight dimensions, to which the 7+1 matrices are related. *Not used:* any formula; the octonionic matrices of the project come from `7+1.nb` and are verified by machine (checks `71-B1`, `71-D6`). The manuscripts say that the relation of the 7+1 matrices to this mathematics is not established in them. *Cited:* manuscripts, Secs. I and II.

4. Statements of the project that rest on these papers

So that a reader can check them against the papers, these are all the statements of the project documents whose only support is one of the nine files:

Statement	Where	Paper
The author's papers define matrices τ^A, $\bar{\tau}^A$, t^A of the kind used here, with the index values 1 to 8, the value 8 corresponding to the value 0 here	M II A; P1 5.2, 6.2	D07, D08
A tilde denotes the transpose in those papers	P1 5.2	D07, D08
D07 exchanges "tau" and "tau bar" and changes the sign of τ^7 relative to an earlier paper	P1 5.2	D07
The construction follows a procedure of Lord	M I	D08, D01
The letters s, t of the 4×4 matrices, self-dual and anti-self-dual, are those of the 1984 paper	M II A (4+4); P1 5.2	D06
The real representations descend from Dirac's representation of the covering group of $SO(3,2)$, through $SO(3,3)$	M I	D02, D03, D06 (and the text of the notebook N04)
The signature 7+1 is the case of the octonions	M I, II	D04, D05, D09

M denotes the two manuscripts (the section numbers are the same in both unless a spacetime is named).

The manuscripts cite about seventy further works. These were not supplied as files; they are cited from the published record, and their bibliographic data are in `manuscript/spacetimes_refs.bib`.

5. Checksums

The following lines of `provenance/inputs_sha256.txt` identify the nine copies (SHA-256; verify with `shasum -a 256 <file>` on macOS):

```
8d2a2c77c7d0fa0a864e9254d465771b6164355b863f969e561eb0677cfcdc90  6dim dirac.pdf
56e95fadec54e9b72e63f9beb0ddf6e424796622a64e7039a21b7ec98fc8fb3f  A Remarkable Representation.pdf
8eddd3a6a63d38531d881e12be56e350cd3f14846c112b53c59f7099ee6a86f1  A new spin-half.pdf
85a8fe5049e60b5b8a65bc531110bc371f96749360c120716b00c0daaebcc0cf  Bryant-octon-17-2-185-txt.pdf
af0befdd8949c78ca55f3c4aac8fb4f741083df9f23415cbd48c875160df64e3  Calibrated geometries.pdf
197d18bfe18917375b11703bbf0b2fe022d97afb41278024841cd0d0ead64d22  Identities satisfied.pdf
27fbbfb90522fecb4f23ec77a23ea621a87b0c9a3ad5d8b29f97dff2cd18d0bc  On the structure of the split
 ↪  octonion algebra.pdf
7be828aad7e05b1753f9b81fdb57f65983a8b38536b49f3bdafeaa16242fc36a  Second gravity.pdf
a436b29fd6b0e0d8ee9819398389619f4427b5384207bd404565ea569dde36c6  Vector cross products.pdf
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